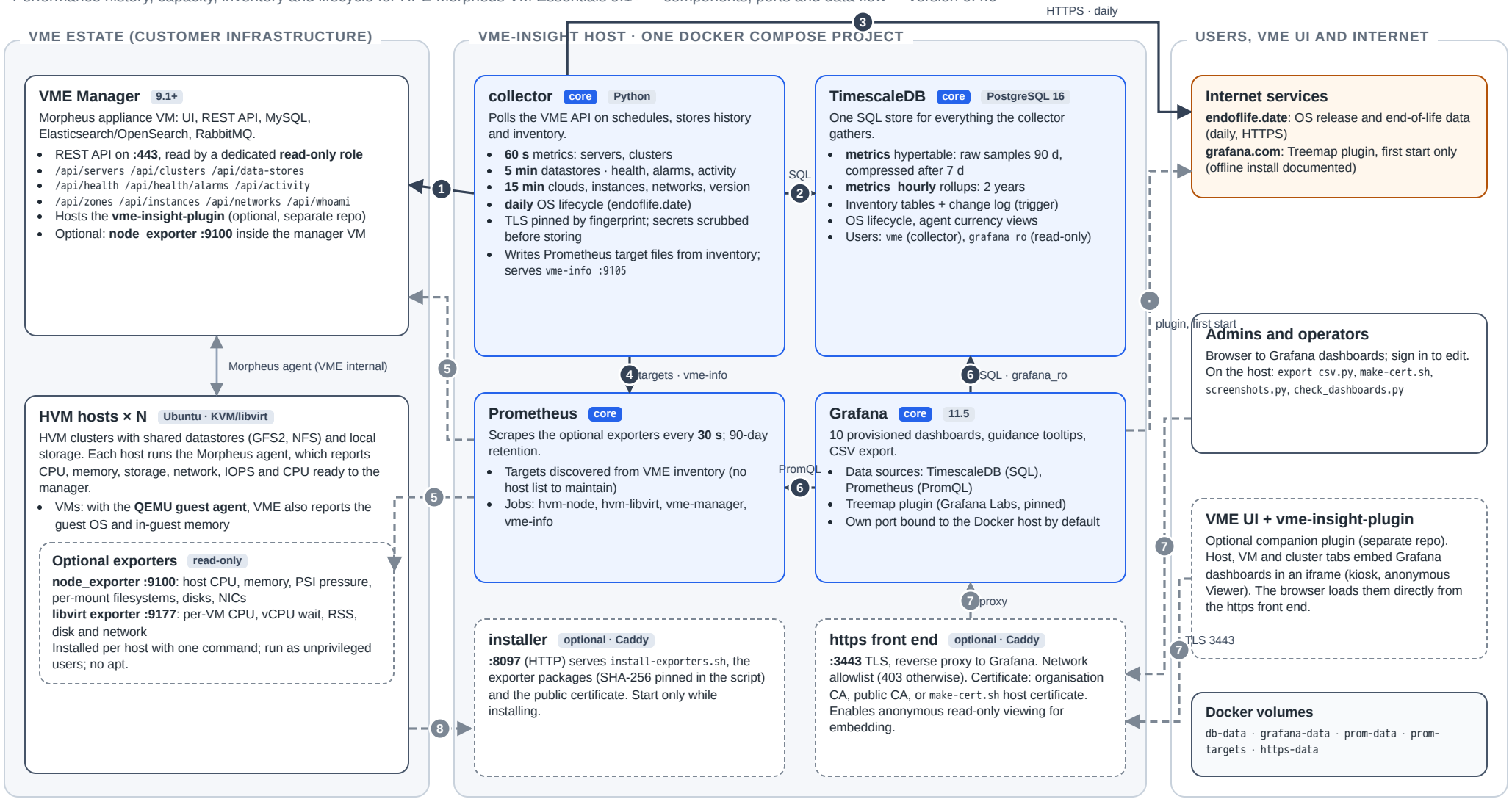
vme-insight architecture

Performance history, capacity, inventory and lifecycle for HPE Morpheus VM Essentials 9.1+ · components, ports and data flow · version 0.4.0



DATA FLOWS

1. **collector → VME API**: HTTPS 443, OAuth password grant (or token), read-only role, certificate pinned. 60 s to daily.
2. **collector → TimescaleDB**: SQL; metrics by bulk COPY, inventory by upsert; change log written by trigger.
3. **collector → endoflife.date**: HTTPS, daily, OS release cycles and EOL dates.
4. **collector → Prometheus**: target files from VME inventory; VME names (datastore, VM) on :9105.
5. **Prometheus → exporters**: HTTP 9100 / 9177 on hosts and manager, every 30 s. Optional.
6. **Grafana → data**: SQL as `grafana_ro` to TimescaleDB; PromQL to Prometheus.
7. **Browser / VME tabs → https front end → Grafana**: TLS 3443, network allowlist. Optional.
8. **Hosts → installer**: one-time HTTP 8097, `curl … | sudo bash`, packages verified by SHA-256.

REQUIRED VS OPTIONAL

Core: collector, TimescaleDB, Prometheus, Grafana. Every API-based dashboard works with only these and a read-only VME account.

Optional: exporters (kernel-level host, VM and filesystem detail), https front end (TLS, allowlist, embedding), installer, VME plugin.

Internet: lifecycle data and the plugin download. Both can be skipped or mirrored.

SECURITY AT A GLANCE

Read-only VME account; http upgraded to https; VME certificate pinned.

Agent API keys, credentials and hashes scrubbed before storage.

Grafana reads as `grafana_ro`; database and Prometheus not published.

Exporters have no auth: bind to management IP, allow only this host.

Browser trust: prefer an organisation CA or a host-only certificate (`CA:FALSE`).

Ports on the vme-insight host: 3000 Grafana (local by default) · 3443 https front end · 8097 installer · internal only: 5432 TimescaleDB, 9090 Prometheus, 9105 vme-info

Community project, not affiliated with HPE · github.com/grahamfairbank/vme-insight